

★ WEEK 16 · PART 1 of 3

Three Families, and How to Choose

Classical, State-Space, and Bayesian - the Whole Course, Sorted

Three questions · three kinds of forecast · one decision-driven choice

The final week opens by stepping back. Everything you learned sorts into three model families, each answering a different question. There is no universal best model - the right one depends on the decision you face.

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PART 1 · THE THREE FAMILIES

One Course, Three Questions

The forecasting competitions that taught the field to test, the three model families and what each one answers, the three kinds of forecast they produce, and how to let the decision choose the model.

The Forecasting Competitions — Part 1

How the field learned to stop arguing and start testing

For years, forecasters argued about which method was best in theory. Then the M-competitions put dozens of methods on the same series and judged them on data they had never seen. The surprise: elaborate models often did NOT beat simple ones.

What the competitions did

- Many methods forecast the very same series.
- Each was graded on held-out future data.
- The scoreboard was public and out-of-sample.

The uncomfortable finding

- Sophistication did not guarantee accuracy.
- Simple methods were often hard to beat.
- Theory alone couldn't settle the argument.

The Forecasting Competitions — Part 2

The lesson: let unseen data decide

The lasting lesson is the backbone of this week. Don't crown a model on elegance or in-sample fit. Fit on the past, then grade every model on the future it never saw - and let that honest test decide.

The lesson

- Out-of-sample accuracy is the real judge.
- In-sample fit can flatter a complex model.
- Compare fairly, on the same held-out data.

How this week uses it

- Apply all three families to one asset.
- Grade them on data none of them saw.
- Let the scoreboard, not taste, decide.

The Whole Course Was One Arc

Three families, three questions

1

Classical asked: what is the mean and variance?

ARMA and GARCH (Weeks 2-6) model the average return and how its volatility clusters - a point forecast and a risk level.

2

State-space asked: what is hidden and changing?

Kalman filters and HMMs (Weeks 8-12) track a moving hidden state or regime behind the noisy data.

3

Bayesian asked: how uncertain is all of it?

Priors and MCMC (Weeks 13-15) turn point estimates into full distributions - honest doubt over states and parameters.

Part 1 in one line: three families answer three different questions - and no single one wins everywhere.

The Three Families at a Glance

Classical models the mean and variance; state-space tracks the hidden regime; Bayesian delivers the full distribution. The pairs thread runs through all three, and out-of-sample testing compares them on the same asset

The Three Families at a Glance

Each family answers a different question about one asset - and the whole course is the arc across them.



Tied together two ways

The pairs-trading thread runs through all three (find the pair, track the hedge ratio, trade with uncertainty), and out-of-sample testing is how we compare them fairly on the very same asset.

Three Families, Three Questions

Why they are complements, not rivals

The families feel like competitors, but they answer different questions. One wants the best single number; one wants to know which hidden regime we are in; one wants the whole range of what could happen. A real decision usually needs one of these more than the others

Different questions

- Classical: the mean and variance of returns.
- State-space: the hidden state or regime.
- Bayesian: the full uncertainty around it all.

So different answers

- A point forecast and a volatility.
- A tracked state or a regime probability.
- A predictive distribution with credible bands.

Classical: ARMA and GARCH

Modeling the mean and the variance of returns

The classical family models returns directly: ARMA captures the predictable part of the mean, and GARCH captures how volatility clusters and persists. Its output is a point forecast of the return and a forecast of its variance.

What it captures

- ARMA: the autocorrelation in the mean.
- GARCH: volatility that clusters over time.
- A parsimonious description of the series.

What it gives

- A point forecast for the next return.
- A forecast of the variance around it.
- Simple, fast, and often hard to beat.

Classical, Across the Course

Where you already used it

The tools you built

- Stationarity and the ACF/PACF (Weeks 1-2).
- AR, MA, ARMA, and ARIMA models (Weeks 3-4).
- GARCH for clustered volatility (Weeks 5-6).

Its strengths and limits

- Strength: clean, interpretable, efficient.
- Limit: assumes fixed parameters.
- Limit: no hidden state, no honest band.

State-Space: Kalman and HMM

Tracking what is hidden and changing

The state-space family assumes something unobserved drives the data - a slowly drifting level, a time-varying beta, a hidden regime. The Kalman filter tracks a continuous hidden state; the HMM tracks a discrete regime. The output is that tracked state.

What it captures

- A hidden state that evolves over time.
- Continuous drift (Kalman) or regimes (HMM).
- Parameters that move, not stay frozen.

What it gives

- A filtered estimate of the hidden state.
- A regime label or probability at each date.
- Adaptation the classical family lacks.

State-Space, Across the Course

Where you already used it

The tools you built

- State-space form and the Kalman filter (Weeks 8-9).
- Hidden Markov models for regimes (Week 10).
- SDEs and short-rate models (Weeks 11-12).

Its strengths and limits

- Strength: tracks moving states and regimes.
- Limit: usually a point estimate of the state.
- Limit: parameters still taken as known.

Bayesian: Full Uncertainty

Parameter doubt and predictive distributions

The Bayesian family stops pretending the parameters are known. It places priors, infers a full posterior, and produces a predictive distribution. Its output is not a line but a fan - the honest range of what could happen, states and parameters included.

What it captures

- Uncertainty about the parameters themselves.
- A joint posterior over states and parameters.
- The two layers of doubt from Week 15.

What it gives

- A predictive distribution, not one number.
- Credible bands on every quantity.
- Risk you can size a decision against.

Bayesian, Across the Course

Where you already used it

The tools you built

- Bayesian inference and conjugacy (Week 13).
- MCMC and HMC-NUTS sampling (Week 14).
- Bayesian state-space and switching (Week 15).

Its strengths and limits

- Strength: honest, decision-ready uncertainty.
- Limit: heavier to compute and to check.
- Limit: needs care with priors and diagnostics.

The Three Families in One Table

The synthesis at a glance

Family	Question it answers	Tools	Forecast
Classical	Mean and variance of returns	ARMA / GARCH	A point + a variance
State-Space	Which hidden state or regime	Kalman / HMM	A tracked state
Bayesian	How uncertain is it all	Priors / MCMC	A full distribution

Read each row as a decision: pick the family whose forecast type matches what your decision needs.

What Each Family Actually Forecasts

Point, regime, or full distribution

The cleanest way to tell the families apart is by the shape of their forecast. Classical gives one number per day. State-space gives the regime you are probably in. Bayesian gives the whole distribution of what could happen next.

Three shapes of answer

- A single line: the point forecast.
- A shaded regime: which state lies ahead.
- A fan: the full predictive distribution.

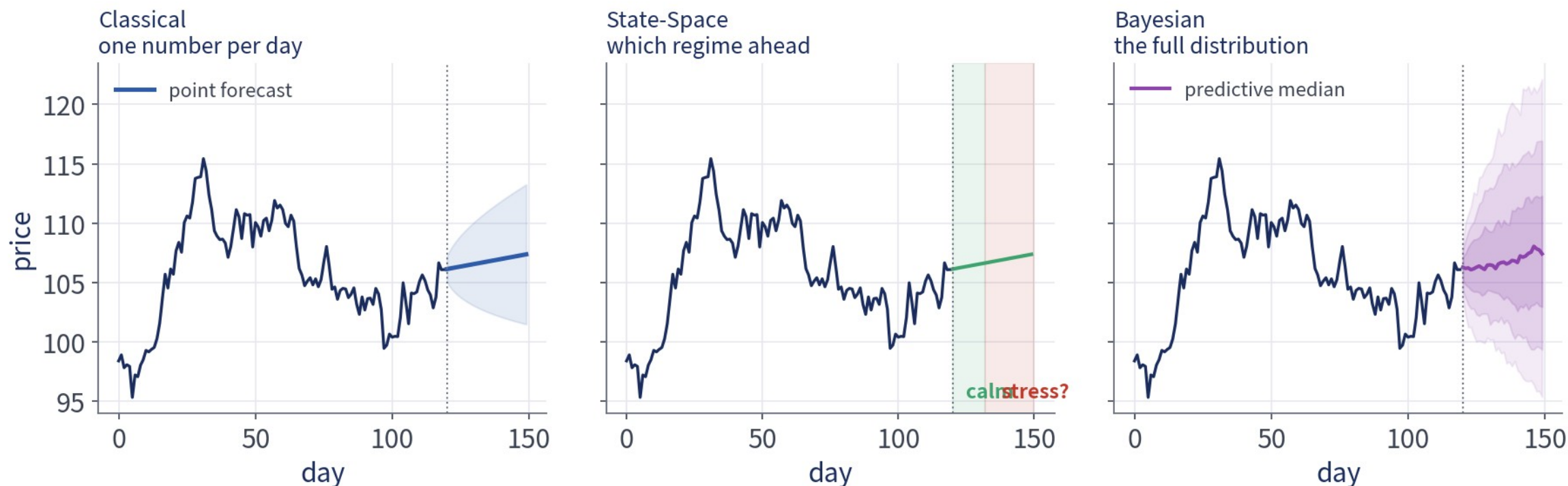
Why the shape matters

- It determines what decisions you can make.
- A number can't tell you the tail risk.
- A distribution can't be beaten for sizing risk.

The Same Asset, Forecast Three Ways

One asset's future as a point forecast (Classical), as a regime call (State-Space), and as a full predictive fan (Bayesian). None is 'the truth' - each answers a different question, and each fits a different decision.

The same asset, forecast three ways - match the forecast to your decision



Match the Forecast to the Decision

The forecast shape you need depends on the choice

If your decision is...

- 'What return do I book?' - you need a point.
- 'Is it safe to be on?' - you need the regime.
- 'How much can I lose?' - you need the distribution.

...then reach for

- Classical, for the single best estimate.
- State-space, for the regime probability.
- Bayesian, for the tail and the credible band.

There Is No Universal Best Model

The right choice depends on the decision

This is the central idea of the week. Asking 'which family is best?' is the wrong question - the competitions showed no model wins everywhere. The right question is: what does THIS decision need? Then the choice follows.

Why no model wins always

- Each is best at its own question.
- Complexity does not guarantee accuracy.
- The data and the decision differ every time.

So how you decide

- Start from the decision, not the model.
- Match the forecast shape to the need.
- When in doubt, run all three and test.

How to Choose in Practice

A short checklist

Ask three questions

- What exactly must the forecast support?
- What shape of answer does that need?
- How much does being wrong cost?

Then act

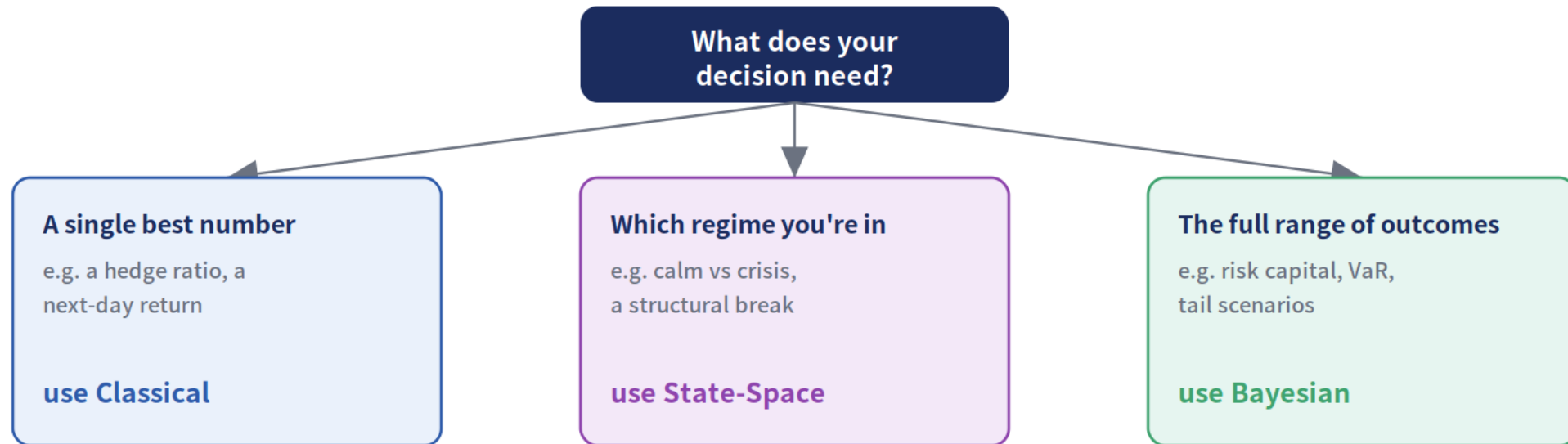
- Pick the family whose output matches.
- If the stakes are high, run all three.
- Let out-of-sample accuracy break ties.

Let the Decision Choose the Model

Ask what the decision needs: a single best number points to Classical; knowing the regime points to State-Space; the full range of outcomes points to Bayesian. And when it matters, run all three and let the test decide

Let the Decision Choose the Model

There is no universal best model. Ask what your decision actually needs, then pick the family.



And often: run all three, then let out-of-sample accuracy break the tie.

The families are complements, not rivals - the capstone compares them fairly on one asset.

The Time-Series Toolbox, in Trading

This is what all the models were for. ARMA sets the entry direction, GARCH the size, cointegration the pair itself, Kalman the hedge ratio, HMM a regime gate, OU the holding period, and Bayesian the conviction. Part 3 assembles them into one trade.

The Time-Series Toolbox, in Trading

Every model you learned maps to a concrete trading job - a signal, a size, a gate, or a holding period.

ARMA (W3-4)

forecasts the mean

-> entry direction: which way the spread moves next

GARCH (W5-6)

forecasts volatility

-> position size: scale inversely with risk; set VaR

Cointegration (W7)

the long-run spread

-> the pair itself: define the spread and its bands

Kalman (W9)

time-varying hedge ratio

-> the long/short ratio, adjusted in real time

HMM (W10)

the regime probability

-> a gate: cut or exit when a crisis regime is likely

OU process (W11)

mean-reversion speed

-> holding period and entry threshold (the half-life)

Bayesian SS (W15)

the credible band

-> conviction sizing: bet less when the band is wide

What Parts 2 and 3 Add

From choosing to testing to delivering

Part 2 - the fair comparison

- Apply all three to one asset.
- Hold-out testing and walk-forward.
- Metrics a trading desk actually uses.

Part 3 - the final project

- The capstone, step by step.
- From comparison to a trading decision.
- Presenting it, and where you go next.

Three Families on One Asset

Direct Claude Code to set up the comparison

GOAL Fit one representative from each family to a single asset and compare their forecast shapes.

[INPUT] Prompt · Claude Code

Write and run `three_families.py` that:

1. loads one asset's returns and splits into a training period and a held-out test period;
2. fits a Classical model (ARMA + GARCH), a State-Space model (Kalman or HMM), and a Bayesian model (a simple posterior/predictive);
3. produces each family's forecast for the test period: a point, a regime, and a distribution;
4. plots all three against the realized path.

Note which decision each output would best serve.

How to work with it

- Same asset, same split.
- Three shapes of forecast.
- None is 'the truth'.
- Part 2 will grade them fairly.

Deliverable: `three_families.py` + a point, a regime, and a distribution forecast for one asset.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1 What question does each of the three families answer?

**Mean/variance,
hidden state, full
uncertainty**

Q2 What forecast shape does each produce?

**A point, a regime, a
distribution**

Q3 How should you choose among them?

**By what the decision
needs, then test**

All three come from the three questions, the three forecast shapes, and the decision-first rule.

Three Misconceptions to Avoid

Where synthesis intuition slips

Common myths

- "The most sophisticated model is the best."
- "A great in-sample fit means great forecasts."
- "One family should be used for everything."

The reality

- The competitions showed simple can win.
- In-sample fit can flatter and mislead.
- Match the family to the decision at hand.

Three Things to Carry Forward

Part 1 in one screen

1

The course sorts into three families

Classical models the mean and variance; state-space tracks the hidden state or regime; Bayesian delivers the full distribution. Each answers a different question about the same asset.

2

The forecast shape is the key difference

A point, a regime, or a full distribution. What you can decide depends on which shape you hold, so match the forecast to the decision you face.

3

There is no universal best model

The competitions taught the field to test, not to argue. Start from the decision, pick the family whose output fits, and when it matters, run all three and let unseen data decide.

Next — Part 2: apply all three to one asset and compare them fairly, out of sample.